

Risk Sharing, International Comovement, and the Goldilocks Trade Elasticity

Ozge Akinici (*FRB - New York*)

Ryan Chahrour (*Cornell University*)

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These results represent the views of the authors and not the views of the Federal Reserve Bank of New York or the Federal Reserve System.

Motivation: International Comovement Puzzles

Backus and Smith (1993):

- Low correlation between relative consumption and real exchange rate.

Kose and Yi (2006):

- Countries that trade more have more correlated business cycles.

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- high home productivity more likely to induce foreign expansion (Drozd et al., 2021)
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High trade elasticity: good for Backus-Smith; bad for intl. transmission

- home and foreign goods are substitutes
- high home productivity less likely to induce foreign expansion
- TOT move little: with NT sector, RER can appreciate (Benigno and Thoenissen, 2008)

Propose *joint* solution to two puzzles

- Key element 1: “just right” trade elasticity
- Key element 2: demand transmission of supply shocks → (Guerrieri et al., 2022)
- Demonstrate in qualitative model
- Quantify in (slightly) more complicated model

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Along the way: explain tension with Corsetti et al. (2008)

The Problem with Low Trade Elasticities

Two-country Armington endowment economy

Income depends on prices & endowments:

$$P_{ht}Y_{ht}$$

$$P_{ft}Y_{ft}$$

Symmetric CES preferences:

$$C_t = \left(a_h^{\frac{1}{\phi}} C_{ht}^{\frac{\phi-1}{\phi}} + a_f^{\frac{1}{\phi}} C_{ft}^{\frac{\phi-1}{\phi}} \right)^{\frac{\phi}{\phi-1}}$$

$$C_t^* = \left(a_f^{\frac{1}{\phi}} C_{ht}^{*\frac{\phi-1}{\phi}} + a_h^{\frac{1}{\phi}} C_{ft}^{*\frac{\phi-1}{\phi}} \right)^{\frac{\phi}{\phi-1}}$$

Demand functions:

$$C_{ft} = a_f \left(\frac{P_{ft}}{P_t} \right)^{-\phi} C_t$$

$$C_{ht}^* = a_f \left(\frac{P_{ht}}{P_t^*} \right)^{-\phi} C_t^*$$

Price indexes:

$$P_t = \left(a_h P_{ht}^{1-\phi} + a_f P_{ft}^{1-\phi} \right)^{\frac{1}{1-\phi}}$$

$$P_t^* = \left(a_h P_{ft}^{1-\phi} + a_f P_{ht}^{1-\phi} \right)^{\frac{1}{1-\phi}}$$

Budgets:

$$P_t C_t = P_{ht} Y_{ht}$$

$$P_t^* C_t^* = P_{ft} Y_{ft}$$

Equilibrium

The final equilibrium condition is balanced trade: $P_{ft}C_{ft} = P_{ht}C_{ht}^*$

Rearranging:

$$\tau \equiv \frac{P_{ft}}{P_{ht}} = \frac{C_{ht}^*(\tau)}{C_{ft}(\tau)}$$

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Defines “best response” terms of trade:

$$\mathcal{T}(\tau) \equiv \tau^{1+\phi} \frac{a_h + a_f \tau^{1-\phi}}{a_f + a_h \tau^{1-\phi}} \frac{Y_f}{Y_h}.$$

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$\mathcal{T}(\tau)$ always upward sloping: higher τ is a higher price of F 's output

- Substitution effect: C_{ht}^* rises
- Wealth effect: C_{ht}^* rises
- Substitution effect: C_{ft} falls
- Wealth effect: C_{ft} falls

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↔ If quantities don't move, then relative price doesn't move.

Steady States: fixed points of $\mathcal{T}(\tau)$

Proposition

Suppose $Y_{ht} = Y_{ft}$. Then there always exists a symmetric equilibrium with

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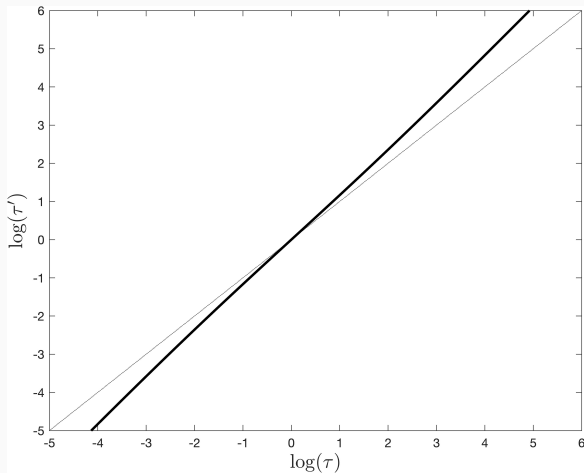
$$\tau = 1.$$

Proposition

Define $\bar{\phi} \equiv 1 - \frac{1}{2a_h}$. When $\phi < \bar{\phi}$, the economy has exactly two additional asymmetric steady-state equilibria, one with $\tau > 1$ and one with $\tau < 1$.

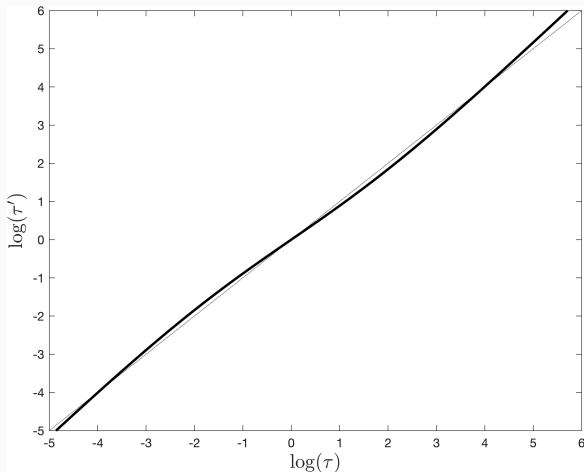
The shape of $\mathcal{T}(\tau)$

For elasticities $\phi > \bar{\phi}$.



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Intuition for multiplicity

Conjecture an increase in P_f .



Foreign country holds higher share of wealth



World consumption basket shifts to f good (home bias)



Further increase in P_f

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Further increase in P_f

If elasticities are low enough, this feedback is self-sustaining

↔ Bodenstein (2010) already knew this!

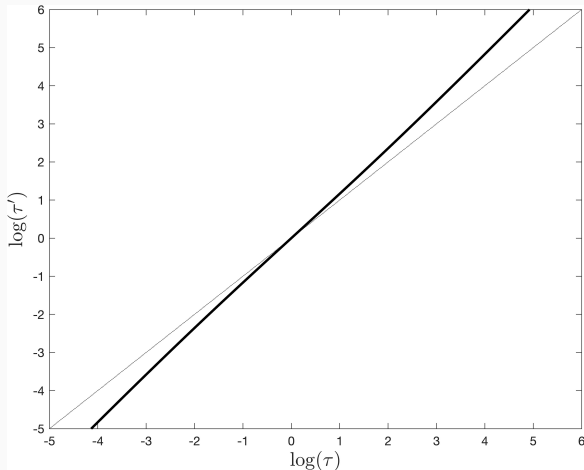
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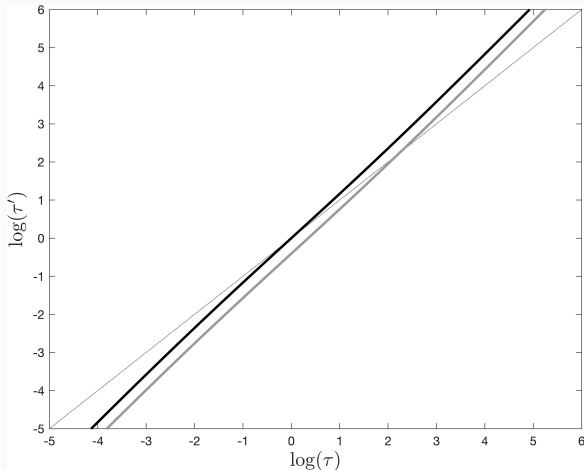
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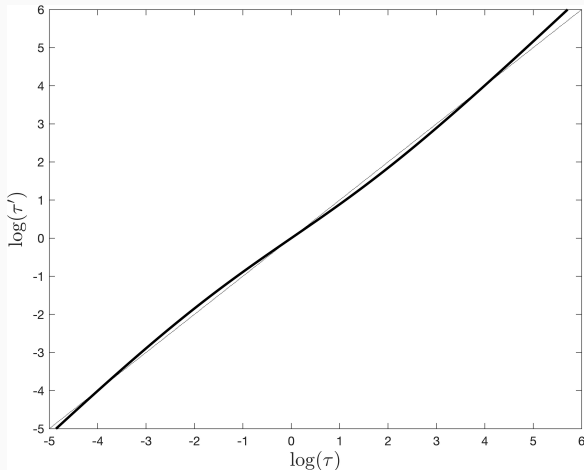
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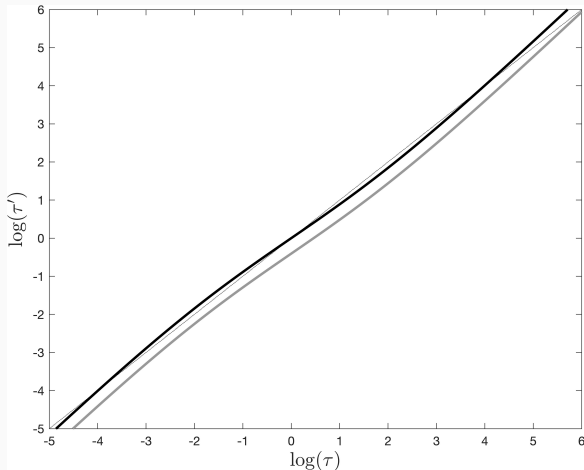
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Proposition

Then:

- *If $\phi > \bar{\phi}$, then an increase in Y_{ht} causes $\tau = P_{ft}/P_{ht}$ to rise.*
- *If $\phi < \bar{\phi}$, then an increase in Y_{ht} causes $\tau = P_{ft}/P_{ht}$ to fall.*

Intuition: “With low elasticity, income effects dominate substitution effects.”

What's stable?

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- Corsetti et al. (2008) reversal relies on perturbation of unstable equilibria
- Stable equilibria give “wrong” consumption-TOT/RER comovements

Qualitative Theory

Minimum changes to model to match two key patterns?

- 1 intermediate value of trade elasticity
- 2 demand-like transmission of supply shocks

Adding production

Firms solve:

$$\max_{N_{ht}} P_{ht} A_{ht} N_{ht}^{\alpha} - W_t N_{ht}.$$

Implies:

$$W_t = P_{ht} A_{ht} \alpha N_{ht}^{\alpha-1}.$$

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Inspired by Guerrieri et al. (2022).

Updated “best response” condition

$$\tilde{\mathcal{T}}(\tau) \equiv \tau^{1+\phi+\alpha/(1-\alpha)} \left(\frac{a_h + a_f \tau^{1-\phi}}{a_f + a_h \tau^{1-\phi}} \right)^\Gamma \left(\frac{A_f}{A_h} \right)^{\frac{1}{1-\alpha}},$$

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Condition for uniqueness less restrictive:

$$\underline{\phi} \equiv \frac{1}{1-\alpha} \left(1 - \frac{1+\alpha}{2a_h} \right) < 1 - \frac{1}{2a_h} = \bar{\phi}.$$

A solution to both puzzles?

Define

$$\tilde{\phi} \equiv \frac{1}{2a_h} > \phi_{\sim} \equiv \frac{a_h - \alpha}{2a_h(1 - \alpha)} > \underline{\phi}$$

Proposition

For any $\phi \in (\phi_{\sim}, \tilde{\phi})$, home traded-good productivity leads to a joint increase in home consumption, home labor, foreign consumption, and foreign labor, with C_{ft} increasing by more than C_{ht} .

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Note: no mention of IES, which is crucial to Guerrieri et al. (2022).

- Low elasticity makes home and foreign good complements
- P_f increases
- Foreign labor demand shifts out
- Sticky wages prevent labor supply response
- Foreign output and consumption rise
- If ϕ is not too low, home consumption rises too.

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- good time to produce at home
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- weak or negative international transmission
- counterfactual correlation between *TOT* and *RER*

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Kekre and Lenel (2024) use demand (β) shocks

- explain other exchange rate patterns
- endowment economy; but what if added production?

A Quantitative Resolution?

Extensions

- 1 Non-traded goods

$$C_t = \left(a_T^{1/\eta} C_{Tt}^{\frac{\eta-1}{\eta}} + a_n^{1/\eta} C_{Nt}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}$$

- 2 Intertemporal preferences

$$E_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\sigma}}{1-\sigma} \right]$$

- 3 Asset trade: home risk-free bond

$$B_{ht} = -B_{ht}^*$$

- 4 Technology $(H \times F) \times (T \times NT)$

$$\Delta \log(A_{fT,t}) = \rho_T \Delta \log(A_{fT,t-1}) + \sigma_t \epsilon_{fT,t}$$

Calibration: Target Moments

Model and data moments.

Moment	Data	Model
$std(\Delta \log(GDP_t))$	1.90	
$corr(\Delta \log(GDP_t), \Delta \log(GDP_{t-1}))$	0.27	
$std(\log(RER_t))/std(\log(GDP_t))$	2.74	
$std(\log(P_N/P_h))/std(\log(GDP_t))$	0.99	
$std(\log(N_t))/std(\log(GDP_t))$	1.36	
$corr(\log(RER_t), \log(C_t/C_t^*))$	0.18	
$corr(\log(RER_t), \log(NX_t))$	-0.15	
$corr(\log(RER_t), \log(TOT_t))$	0.31	
$corr(\log(GDP_t), \log(GDP_t^*))$	0.77	
$corr(\log(C_t), \log(C_t^*))$	0.69	
$corr(\log(N_t), \log(N_t^*))$	0.55	

Note: All moments based on annual data for the US relative to the OECD, from 1970 - 2019. Non-growth rate data is HP-filtered log-levels and penalty 6.25.

Calibration: Parameters

Model calibration.

Description	Parameter	Value
Discount factor	β	0.960
Inverse IES	σ	1.000
Labor share	α	0.720
Nontraded share	a_n	0.700
Home traded share	a_h	0.600
T-NT elasticity	η	
Trade elasticity	ϕ	
Import adjustment cost	ϑ	
AR of traded tech	ρ_t	
AR of nontraded tech	ρ_n	
SD of traded tech	σ_t	
SD of nontraded tech	σ_n	

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Home traded share	a_h	0.600
T-NT elasticity	η	0.467
Trade elasticity	ϕ	0.737
Import adjustment cost	ϑ	0.211
AR of traded tech	ρ_t	0.676
AR of nontraded tech	ρ_n	0.000
SD of traded tech	σ_t	1.107
SD of nontraded tech	σ_n	0.846

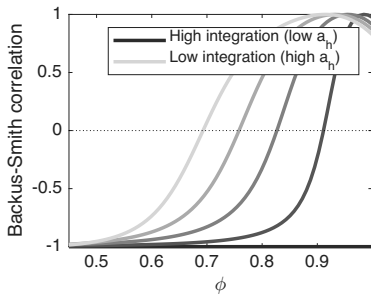
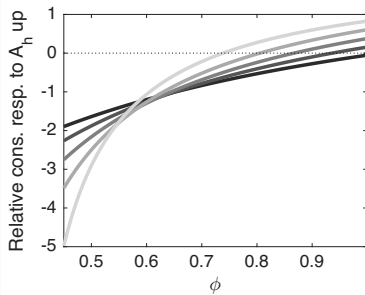
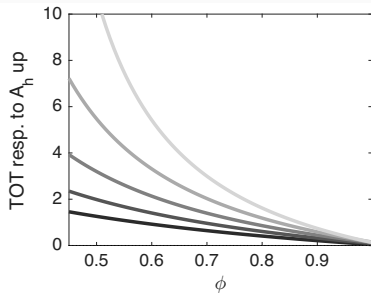
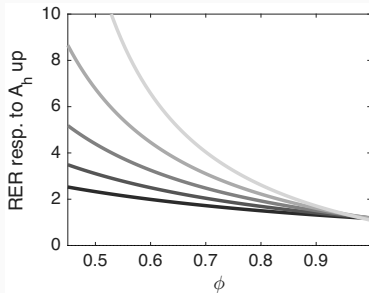
Calibration: Model vs. Data

Model and data moments.

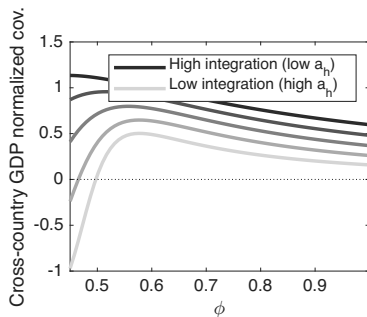
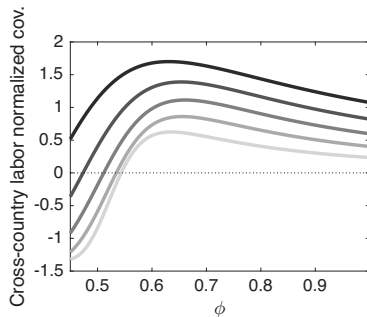
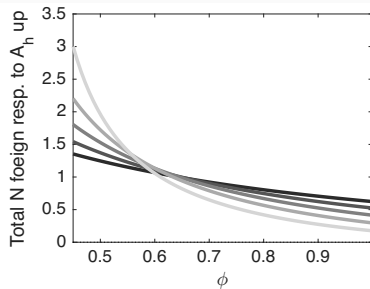
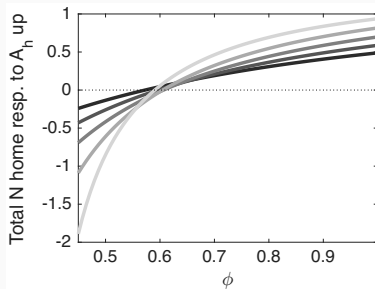
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$corr(\log(RER_t), \log(TOT_t))$	0.31	0.86
$corr(\log(GDP_t), \log(GDP_t^*))$	0.77	0.72
$corr(\log(C_t), \log(C_t^*))$	0.69	0.70
$corr(\log(N_t), \log(N_t^*))$	0.55	0.76

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Backus-Smith comparative statics



Trade-comovement comparative statics



- 1 Intuitive tension in resolving two old puzzles
- 2 One possible resolution
- 3 Future directions
 - Adding capital
 - Allowing for anticipated disturbances

References

- Backus, D. K. and G. W. Smith (1993). Consumption and real exchange rates in dynamic economies with non-traded goods. *Journal of International Economics* 35(3-4), 297–316.
- Benigno, G. and C. Thoenissen (2008). Consumption and real exchange rates with incomplete markets and non-traded goods. *Journal of International Money and Finance* 27(6), 926–948.
- Bodenstein, M. (2010). Trade elasticity of substitution and equilibrium dynamics. *Journal of Economic Theory* 145(3), 1033–1059.
- Corsetti, G., L. Dedola, and S. Leduc (2008). International risk sharing and the transmission of productivity shocks. *Review of Economic Studies* 75, 443–473.
- Drozdz, L. A., S. Kolbin, and J. B. Nosal (2021). The trade-comovement puzzle. *American Economic Journal: Macroeconomics* 13(2), 78–120.
- Guerrieri, V., G. Lorenzoni, L. Straub, and I. Werning (2022). Macroeconomic implications of covid-19: Can negative supply shocks cause demand shortages? *American Economic Review* 112(5), 1437–1474.
- Kekre, R. and M. Lenel (2024). Exchange rates, natural rates, and the price of risk. *University of Chicago, Becker Friedman Institute for Economics Working Paper* (2024-114).
- Kose, M. A. and K.-M. Yi (2006). Can the standard international business cycle model explain the relation between trade and comovement? *Journal of International Economics* 68(2), 267–295.